Colon Cancer Screening

Epidemiology of Colorectal Cancer

- 3rd most commonly diagnosed cancer and the 2nd leading cause of cancer death in the US
- Estimated ~154,270 new diagnoses and ~52,900 deaths in 2025
- Incidence 33% higher in men (41.5 per 100,000) than women (31.2 per 100,000)
- Overall incidence declined 0.9% annually (2013–2022), driven by decreases in adults 65 years
- Lifetime risk of CRC in the US: approximately 4%–5%

(**siegele70067?**)

Early-onset Colorectal cancer

- Incidence increasing ~2%–3% annually in adults aged 20–49 years
- CRC is now the leading cause of cancer death in men 50 years
- 20% (1 in 5) of CRCs in 2019 were in patients 54 years, up from 11% (1 in 10) in 1995

(**jayakrishnan1373?**)

Non-modifiable Risk Factors for Colorectal Cancer

- Age (incidence rises sharply after age 50)
- Male sex

- Race/ethnicity (highest in Alaska Native, American Indian, Black populations)
- Family history of CRC or advanced adenomas in first-degree relatives
- Personal history of adenomas, SSP/SSLs, or IBD
- Hereditary syndromes
    - Lynch Syndrome
    - Familial Adenomatous Polyposis

Modifiable Risk Factors for Colorectal Cancer

- Obesity and excess body weight
- Physical inactivity
- Tobacco smoking
- High alcohol consumption
- Diet high in red/processed meat, low in fruits/vegetables/fiber
- Over 50% of CRC cases are attributable to modifiable lifestyle factors[9][10]

The Adenoma-Carcinoma Sequence

Most CRCs develop over 10–15 years from precursor polyps through a stepwise accumulation of genetic and epigenetic alterations[11][12] Two Major Pathways:

1. Conventional Adenoma-Carcinoma Pathway (70%–90% of CRCs)[3][12] - Initiated by APC gene mutation → KRAS activation → TP53 inactivation - Characterized by chromosomal instability (CIN) - Produces microsatellite-stable tumors
2. Serrated Neoplasia Pathway (10%–20% of CRCs)[12][13] - Arises from sessile serrated lesions (SSLs) - Driven by BRAF mutations and CpG island methylator phenotype (CIMP) - Often leads to microsatellite instability (MSI-H) via MLH1 promoter methylation - Predominates in proximal colon, elderly, female patients[11] Clinical Implication:

Screening for Colorectal Cancer: US Preventive Services Task Force Recommendation Statement. US Preventive Services Task Force, Davidson KW, Barry MJ, et al. JAMA. 2021;325(19):1965-1977. doi:10.1001/jama.2021.6238.
Colorectal Cancer Screening for Average-Risk Adults: 2018 Guideline Update From the American Cancer Society. Wolf AMD, Fontham ETH, Church TR, et al. CA: A Cancer Journal for Clinicians. 2018;68(4):250-281. doi:10.3322/caac.21457.

The prolonged adenoma-to-carcinoma timeline provides a window for screening to detect and remove precancerous lesions before malignant transformation[6][14]

Risk Stratification: Average vs. Increased Risk

Average Risk (per NCCN) — ALL of the following:[8] - Age 45–75 years - No personal history of adenomas, serrated sessile lesions (SSL), CRC, IBD, cystic fibrosis, or childhood cancer - No hereditary CRC syndromes - No family history of CRC or advanced adenomas in first-degree relatives Increased Risk — ANY of the following:[8] - Personal history of adenomas or SSP/SSLs - Personal history of CRC or IBD - Hereditary syndromes (Lynch, FAP, etc.) - 1 first-degree relative with CRC or advanced adenoma - Personal history of cystic fibrosis or childhood cancer Key Principle: Average-risk individuals may choose any recommended screening modality. Increased-risk individuals should undergo colonoscopy as the preferred method[8]

When to Start and Stop Screening

Age to Begin Screening (Average Risk): - USPSTF: Age 45 (Grade B for 45–49; Grade A for 50–75)[9] - ACS: Age 45[10][15] - NCCN: Age 45 (data stronger for age 50, but lower-level evidence supports 45)[8] - ACG: Age 45 (conditional); age 50 (strong recommendation)[13][16] Rationale for Lowering to Age 45: - Rising early-onset CRC incidence[7] - Modeling: 22–27 additional life-years gained per 1,000 individuals screened starting at 45 vs. 50[7] When to Stop: - Screening recommended through age 75[8][9] - Ages 76–85: Individualize based on comorbidities, life expectancy, and prior screening history[8] - Requires estimated life expectancy 10 years[8]

Screening Modalities Overview

Stool-Based Tests (Two-Step: positive result → colonoscopy):

- Guaiac-based FOBT (gFOBT) — annually

- Fecal immunochemical test (FIT) — annually
- Multi-target stool DNA (mt-sDNA, e.g., Cologuard) — every 3 years
- Multi-target stool RNA (mt-sRNA) — every 3 years[8]

Direct Visualization Tests:

- Colonoscopy — every 10 years
- Flexible sigmoidoscopy — every 5 years (± annual FIT)
- CT colonography — every 5 years[8][9]

Key Points:

- Colonoscopy is the only one-step modality (diagnostic + therapeutic)[17]
- FIT has replaced gFOBT in most settings due to superior sensitivity, no dietary restrictions, and single-sample convenience[5][8]
- All positive stool-based tests require follow-up colonoscopy as soon as possible and no later than 9 months[8]

Comparative Performance of Screening Tests

Test	CRC Sensitivity	Adenoma Sensitivity	Specificity	Interval
Colonoscopy	95%	75-95% (>6mm)	-	10 yrs
FIT	74%	23-25%	94%	Annual
Cologuard	94%	46%	-	3 years
FOBT	50-75%	6-21%	96-98%	Annual

FIT has replaced guaiac-based Fecal Occult Blood (FOBT)

Data from NCCN Colorectal Cancer Screening Guidelines (v2.2026)

Comparative Performance of Screening Tests

SLIDE 9 — The NordICC Randomized Trial of Screening Colonoscopy

- 84,585 participants aged 55–64 in Poland, Norway, Sweden
- Randomized 1:2 to invitation for colonoscopy vs. usual care
- Only 42% of invited participants underwent colonoscopy

10-year results (intention to screen):

- CRC incidence: 0.98% vs. 1.20% (RR 0.82; 95% CI 0.70–0.93)
- CRC mortality: 0.28% vs. 0.31% (RR 0.90; 95% CI 0.64–1.16) — not significant
- NNI (number needed to invite) to prevent 1 CRC case: 45

SLIDE 9 — The NordICC Randomized Trial of Screening Colonoscopy

- 84,585 participants aged 55–64 in Poland, Norway, Sweden
- Randomized 1:2 to invitation for colonoscopy vs. usual care
- Only 42% of invited participants underwent colonoscopy

Per-Protocol Analysis (those actually screened):

- CRC incidence reduced by 31%–41%
- CRC mortality reduced by up to 50%

SLIDE 10 — Increased-Risk Screening: Family History

Per NCCN Guidelines:[8] Family History Start Screening Modality Surveillance 1 FDR with CRC at any age Age 40 or 10 years before earliest CRC diagnosis (whichever first) Colonoscopy Every 5 years 2nd/3rd-degree relatives with CRC Age 45 Colonoscopy Every 10 years FDR with advanced

16. Effect of Colonoscopy Screening on Risks of Colorectal Cancer and Related Death. Bretthauer M, Løberg M, Wieszczy P, et al. The New England Journal of Medicine. 2022;387(17):1547-1556. doi:10.1056/NEJMoa2208375.

17. Long-Term Effects of Colonoscopy Screening on Colorectal Cancer Incidence and Mortality: A Multicountry, Population-Based Randomised Controlled Trial. Kaminski MF, Kalager M, Løberg M, et al. Lancet (London, England). 2026;:S0140-6736(26)00508-8. doi:10.1016/S0140-6736(26)00508-8.

16. Effect of Colonoscopy Screening on Risks of Colorectal Cancer and Related Death. Bretthauer M, Løberg M, Wieszczy P, et al. The New England Journal of Medicine. 2022;387(17):1547-1556. doi:10.1056/NEJMoa2208375.

17. Long-Term Effects of Colonoscopy Screening on Colorectal Cancer Incidence and Mortality: A Multicountry, Population-Based Randomised Controlled Trial. Kaminski MF, Kalager M, Løberg M, et al. Lancet (London, England). 2026;:S0140-6736(26)00508-8. doi:10.1016/S0140-6736(26)00508-8.

adenoma or advanced SSP/SSL Age 40 or age of onset in relative (whichever first) Colonoscopy Every 5–10 years FDR = first-degree relative Important: If criteria for a hereditary cancer syndrome are met (e.g., Lynch, FAP), refer for genetic evaluation and follow syndrome-specific guidelines[8]

SLIDE 11 — Post-Polypectomy Surveillance

Surveillance intervals assume complete resection, adequate bowel preparation, and complete examination[8] General Principles:

- Low-risk adenomas or SSP/SSLs: May not have substantially increased risk of metachronous advanced neoplasia compared to general population
- High-risk adenomas or SSP/SSLs with dysplasia: Surveillance interval should not exceed 5 years
- 10 cumulative adenomas: Consider germline testing for polyposis syndromes
- 20 serrated polyps: Consider serrated polyposis syndrome[8]

Histologic Considerations:

- SSP/SSLs without dysplasia are managed like tubular adenomas[8]
- SSP/SSLs with dysplasia are managed like high-risk adenomas[8]
- Large (1 cm) hyperplastic polyps are often reclassified as SSLs on expert review[8]

SLIDE 12 — Special Populations Inflammatory Bowel Disease (IBD):[8] - Ulcerative colitis and Crohn's colitis confer increased CRC risk - Colonoscopy is the preferred screening method - Follow IBD-specific

References 1. Colorectal Cancer Statistics, 2026. Siegel RL, Wagle NS, Star J, et al. CA: A Cancer Journal for Clinicians. 2026 Mar-Apr;76(2):e70067. doi:10.3322/caac.70067. 2. Colorectal Cancer Statistics, 2023. Siegel RL, Wagle NS, Cercek A, Smith RA, Jemal A. CA: A Cancer Journal for Clinicians. 2023 May-Jun;73(3):233-254. doi:10.3322/caac.21772. 3. Aspirin and Other Nonsteroidal Anti-Inflammatory Drugs (NSAIDs) for Preventing Colorectal Cancer and Colorectal Adenoma in the General Population. Cai Z, Meng Y, Yang W, et al. The Cochrane Database of Systematic Reviews. 2026;4:CD015266. doi:10.1002/14651858.CD015266.pub3. 4. Colorectal Cancer Facts & Figures. Angela Giaquinto, Tyler Kratzer, Adair Minihan, et al. American Cancer Society (2025).

5. AGA Clinical Practice Update on Approach to the Use of Noninvasive Colorectal Cancer Screening Options: Commentary. Burke CA, Lieberman D, Feuerstein JD. Gastroenterology. 2022;162(3):952-956. doi:10.1053/j.gastro.2021.09.075. 6. Colorectal Adenomas. Strum WB. The New England Journal of Medicine. 2016;374(11):1065-75. doi:10.1056/NEJMra1513581.
6. Early-Onset Gastrointestinal Cancers. Jayakrishnan T, Ng K. JAMA. 2025;334(15):1373-1385. doi:10.1001/jama.2025.10218.
7. Colorectal Cancer Screening. National Comprehensive Cancer Network. Updated 2026-04-23.
8. Screening for Colorectal Cancer: US Preventive Services Task Force Recommendation Statement. US Preventive Services Task Force, Davidson KW, Barry MJ, et al. JAMA. 2021;325(19):1965-1977. doi:10.1001/jama.2021.6238.
9. Colorectal Cancer Screening for Average-Risk Adults: 2018 Guideline Update From the American Cancer Society. Wolf AMD, Fontham ETH, Church TR, et al. CA: A Cancer Journal for Clinicians. 2018;68(4):250-281. doi:10.3322/caac.21457.
10. Colorectal Cancer. Brenner H, Kloor M, Pox CP. Lancet (London, England). 2014;383(9927):1490-1502. doi:10.1016/S0140-6736(13)61649-9.
11. Colorectal Cancer. Dekker E, Tanis PJ, Vleugels JLA, Kasi PM, Wallace MB. Lancet (London, Eng-

land). 2019;394(10207):1467-1480. doi:10.1016/S0140-6736(19)32319-0.

12. ACG Clinical Guidelines: Colorectal Cancer Screening 2021. Shaukat A, Kahi CJ, Burke CA, et al. The American Journal of Gastroenterology. 2021;116(3):458-479. doi:10.14309/ajg.0000000000001122.

13. The Biology of Colorectal Cancer. Implications for Pretreatment and Follow-Up Management. Boland CR. Cancer. 1993;71(12 Suppl):4180-6. doi:10.1002/1097-0142(19930615)71:12+<4180::aid-cncr2820711804>3.0.co;2-g.

14. Cancer Prevention and Early Detection Facts & Figures. Rick Alteri, Deana Baptiste, Emily Butler Bell, et al. American Cancer Society (2025).

15. AGA Clinical Practice Update on Risk Stratification for Colorectal Cancer Screening and Post-Polypectomy Surveillance: Expert Review. Issaka RB, Chan AT, Gupta S. Gastroenterology. 2023;165(5):1280-1291. doi:10.1053/j.gastro.2023.06.033.

16. Colorectal Cancer Screening: Updated Guidelines From the American College of Gastroenterology. de Kanter C, Dhaliwal S, Hawks M. American Family Physician. 2022;105(3):327-329.

17. Effect of Colonoscopy Screening on Risks of Colorectal Cancer and Related Death. Bretthauer M, Løberg M, Wieszczy P, et al. The New England Journal of Medicine. 2022;387(17):1547-1556. doi:10.1056/NEJMoa2208375.

18. Long-Term Effects of Colonoscopy Screening on Colorectal Cancer Incidence and Mortality: A Multicountry, Population-Based Randomised Controlled Trial. Kaminski MF, Kalager M, Løberg M, et al. Lancet (London, England). 2026;:S0140-6736(26)00508-8. doi:10.1016/S0140-6736(26)00508-8.

19. Effect of Colonoscopy Screening on Risks of Colorectal Cancer and Related Death: Instrumental Variable Estimation of Per-Protocol Effects. Shi J, Løberg M, Kalager M,

et al. European Journal of Epidemiology. 2025;40(4):419-425. doi:10.1007/s10654-025-01209-w.